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Exploring the drivers of users' adoption of museum digital humans

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Digital humans are emerging as innovative media for interpreting the cultural heritage value of museums. Based on the Technology Acceptance Model, this study examines how information richness, information quality, aesthetic experience, and flow experience influence users' behavioural intention toward digital humans. Using Partial Least Squares Structural Equation Modelling, 265 valid responses were collected from participants who interacted with the National Museum of China's digital human Ai Wenwen through the "Ai Kan Wenwu" video series. Results indicate that all hypotheses were supported except for the effect of information quality on flow experience. Specifically, information richness significantly enhances information quality and aesthetic experience, while aesthetic experience influences both perceived usefulness and ease of use, and further promotes usage intention through flow experience. By introducing a cognitive–affective dual-path mechanism, this study extends TAM's applicability to cultural heritage narrative contexts and highlights the potential of digital humans to evoke cognitive, emotional, and immersive engagement.

With the empowerment of computer graphics, human–computer interaction, virtual reality, and artificial-intelligence technologies, the interpretation and display of cultural heritage values have entered a digitally driven stage, prompting a profound transformation in how heritage resources are revitalised and utilised. Compared with traditional exhibitions, digital presentation compensates for limitations such as spatial constraints, temporal discontinuity, monotony of form, and superficial value explanation. It offers distinct advantages for heritage conservation, value communication, public education, and visitor experience.

Against this backdrop, an essential question arises: what do audiences expect to gain from engaging with cultural heritage—is the historical connotation itself paramount, or the digital form through which it is mediated[1]? This question reveals that mechanisms of value interpretation within heritage digitisation remain immature, as a gap still persists between the surface content displayed and the deeper cultural meanings intended. Optimising narrative strategies and dissemination pathways has therefore become imperative.

In response, scholars have increasingly focused on digital storytelling mechanisms grounded in heritage values. Prior studies suggest that digital narratives not only expose the deep value connections beyond the material form of heritage but also facilitate the construction of visitors' cultural cognition[2]. Moreover, they promote positive emotional engagement and value identification during behavioural transformation[3]. As museums shift from an object-centred to an experience-centred paradigm[4], digital humans are being widely employed to deliver interactive heritage-narrative experiences[5]. They serve not only as narrators within exhibitions but also as vehicles for performing intangible-heritage content such as traditional crafts, thereby enhancing accessibility and educational value[6].

A review of existing literature reveals that research on user experiences with museum digital humans remains fragmented, which can be summarised in three main directions: (1) Some scholars have explored the roles of digital humans in explanation, guidance, and interactive engagement, showing that anthropomorphic design and affective interaction enhance immersion and enjoyment[5,7]. (2) Other studies focus on technological or design attributes—such as speech recognition, visual appearance, interaction fluency, and aesthetic appeal—and their influence on user experience[8–10]. (3) A smaller body of work considers the effects of digital humans on learning outcomes and information-acquisition efficiency[11,12]. Although these studies provide valuable insights, they fail to address a fundamental question: what are the core factors driving users to accept and continuously engage with museum digital humans? This theoretical gap leaves museums without a systematic framework to guide narrative design and experience optimisation when investing in digital-human development.

To address this issue, the classic Technology Acceptance Model (TAM) offers a logically rigorous starting point, where the constructs of perceived usefulness and perceived ease of use are central. However, it must be recognised that TAM was originally designed to explain instrumental software adoption in workplace settings aimed at improving productivity[13].

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When directly applied to emotionally engaging, experience-oriented cultural media such as museum digital humans, TAM's theoretical limitations become apparent: it cannot adequately account for the affective, aesthetic, and immersive experiences that constitute the distinctive appeal of digital humans compared with ordinary information systems.

Accordingly, this study identifies two major research gaps:

Gap 1: Limited theoretical perspective. Current research lacks an integrative model that simultaneously encompasses users' cognitive judgements (e.g., usefulness, usability) and affective experiences (e.g., pleasure, immersion) to systematically reveal how intention to use develops.

Gap 2: Omission of core experiential dimensions. The aesthetic allure unique to digital humans—and its potential to elicit deep immersive states—has not yet been empirically tested within a unified theoretical framework.

To bridge these gaps, the present study moves beyond merely adding variables to TAM; instead, it undertakes a theoretical expansion. We propose a cognitive–affective dual-path model, which retains TAM's cognitive pathway while introducing an affective pathway triggered by aesthetic experience and mediated through flow experience to influence behavioural intention. This extension repositions TAM from a "tool-efficiency" model to an "experience-acceptance" model in the context of heritage-digital-human use, providing a more holistic lens through which to understand the complex interactions between users and virtual agents.

Using the case of Ai Wenwen at the National Museum of China, this study employs survey data and structural-equation modelling to address the following questions:

RQ1: What are the key mechanisms that drive users' sustained intention to engage with museum digital humans? Specifically, how do the cognitive pathway (represented by perceived usefulness and ease of use) and the affective pathway (centred on aesthetic and flow experience) interact?

RQ2: What role does aesthetic experience play within the model? Does it, as theoretically expected, influence both cognitive evaluation and emotional immersion?

RQ3: Does the proposed extended model outperform the classic TAM in explaining users' behavioural intention?

By answering these questions, the study aims not only to offer empirical evidence for optimising digital-human design in museums but also to advance TAM theory within the broader context of the experience economy and digital heritage communication.

Methods

Research overview

The deep integration of digital technologies is systematically reshaping the paradigms through which cultural heritage is interpreted. At the core of this transformation lies the enhancement of interactivity and narrativity, which together reconstruct the connection between users and heritage. Collaborative applications of augmented reality (AR), virtual reality (VR), and interactive storytelling have been shown to heighten immersive experience, facilitate cultural-knowledge transfer, and foster local identity[14,15]. For instance, design-thinking-driven digital installations strengthen the transmission of historical information through sensory engagement[16], while VR-based serious games provide new multimodal means of interpreting the colours and symbolism of heritage artefacts[17]. Collectively, these practices point to a shift from one-way information delivery towards participatory, affective "co-creation of experience" [18,19].

Within this landscape, digital humans—embodied, personified interfaces—have become cutting-edge vehicles that integrate the aforementioned technological advantages to deepen the interpretation of heritage value. Research demonstrates that digital humans can employ diverse narrative forms such as legends, poetry, and dance to heighten user immersion and enhance accessibility to exhibits within virtual heritage environments[5]. Their value lies not only in cost-effective audience engagement[7] but also in functioning as empathetic media that, through story-driven presentation, evoke renewed understanding and emotional resonance with cultural heritage[20].

As digital technology continues to advance, museum exhibition forms are evolving. Studies have observed that digital and virtual interactions are reconstructing the "human–artefact–space" relationship[21], rendering information structures more flexible and plastic[22,23]. In this broader trend, the digital human—as both interface and narrative medium—has the potential to participate in and reinforce this relational reconstruction, becoming a crucial node in the process of generating cultural meaning.

Yet despite widespread recognition of their capacity to enhance engagement, comprehension, and emotional impact[11], most existing studies remain confined to technical implementation, functional description, or qualitative analysis of user experience. A key unanswered question persists: what psychological mechanisms drive users to accept and continually use museum digital humans? When users expect an experience that is simultaneously entertaining, aesthetic, and educational[24], how do aesthetic enjoyment and deep immersion interact with utilitarian cognition to foster usage intention and value identification? Without a consolidated framework to guide narrative design and experience optimisation, museums struggle to achieve the ultimate goal of "moving people emotionally" while communicating cultural value.

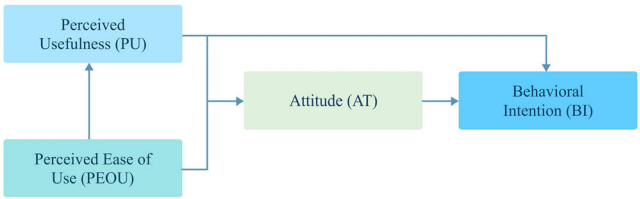

Fig. 1 | Technology acceptance model (TAM). The model illustrates a cognition–behaviour pathway in which perceived usefulness and perceived ease of use jointly shape users' behavioural intention toward technology adoption.

Technology Acceptance Model

The Technology Acceptance Model (TAM), proposed by Davis (1989; see Fig. 1), is one of the most influential frameworks for explaining user behaviour in the field of information systems. Its central logic posits that an individual's intention to use a technology is driven by cognitive evaluations, primarily perceived usefulness (PU) and perceived ease of use (PEOU). Perceived ease of use positively affects perceived usefulness, and together they shape behavioural intention, forming a cognition–attitude–behaviour path[13]. In essence, when users believe a system helps them achieve goals and is easy to operate, adoption and continued use are more likely.

In the domains of museums and heritage communication, TAM has been widely employed to evaluate public acceptance and use intention toward emerging digital technologies[25–27]. For instance, user attitudes and adoption intentions in digital-museum settings can be effectively predicted through TAM[25]; in mixed-reality guide systems, both perceived usefulness and ease of use have been identified as key determinants of behavioural intention[28]. These findings affirm that TAM provides a robust theoretical basis for understanding human–computer interaction within cultural contexts.

A notable strength of TAM lies in its extensibility and contextual adaptability. Because the model does not rigidly constrain external variables, researchers can incorporate new belief constructs or moderating factors suited to specific research contexts, thereby enhancing explanatory and predictive power[29,30]. Consequently, TAM has become a highly versatile analytical framework for heritage digitalisation research. However, its original purpose—to explain the adoption of instrumental, workplace-oriented systems—limits its ability to account for experiential and emotional technologies. As human–computer interaction evolves, TAM applications have gradually shifted from functional technologies to experiential media. Scholars now recognise that beyond cognitive judgement, affective factors such as aesthetic pleasure, emotional involvement, and immersion play crucial roles in user adoption, with proven explanatory potential in virtual-reality, metaverse, and digital-culture settings[31–33]. Accordingly, incorporating affective-driven mechanisms alongside the traditional "usefulness–ease-of-use" path reveals the authentic psychological responses of users within highly immersive environments.

Information richness

Information Richness Theory[34–36] defines information richness as a medium's capacity to change the receiver's understanding within a given time frame. Its key attributes include feedback immediacy, multiplicity of cues, semantic variety, and personalisation[37]. As a quintessential multimodal interface, a digital human conveys messages through synchronised voice, expression, gesture, and virtual environment, thereby transcending the limits of text or unidirectional audio.

Empirical evidence shows that information richness significantly influences perceived information quality, especially in online and virtual settings where users' cognitive judgements and experience are highly sensitive to multimodal stimuli[38,39]. Rich and well-structured visual presentation enhances cognitive fluency and exploratory value, which in turn shape aesthetic preference and acceptance[40]. Within an organised yet open cognitive framework, users decode and reconstruct informational value more effectively[41].

This mechanism corresponds closely with that of museum digital humans: through embodied multimodal storytelling, they present intricate cultural-heritage details that intensify both cognitive engagement and immersive experience. Hence, information richness constitutes the core capability that animates heritage content and converts static artefacts into dynamic narrative forms.

H1: Information richness positively influences perceived information quality of cultural heritage museum digital humans.

H2: Information richness positively influences the aesthetic experience of cultural heritage museum digital humans.

Information quality

Information quality refers to users' evaluation of the accuracy and adequacy of information delivered by a system[42]. Weber[43] identifies its principal characteristics as authenticity, accuracy, completeness, uniqueness (non-redundancy), timeliness, relevance, comprehensibility, precision, conciseness, and informational volume[43]. For museum digital humans entrusted with knowledge transmission and heritage interpretation, information quality underpins their credibility as "virtual experts" and sustains users' cognitive immersion. The authenticity, accuracy, and logical coherence of the multimodal content directly influence perceived information quality, which in turn positively affects perceptions of usefulness and ease of use[44–46].

High-quality informational output enhances experience via two routes. First, accurate and coherent content reduces users' cognitive load when processing complex cultural concepts, thereby improving perceived ease of use[47]. Second, reliable and in-depth information stimulates performance motivation and heightens perceived usefulness[48]. Furthermore, although mobile interfaces impose display constraints, engaging and well-presented information can still increase attention and enjoyment, reinforcing immersion and flow[49]. Information quality thus activates and sustains the cognitive pathway, establishing a trustworthy foundation for subsequent affective and immersive experiences.

H3: Information quality positively influences the perceived usefulness of cultural heritage museum digital humans.

H4: Information quality positively influences perceived ease of use of cultural heritage museum digital humans.

H5: Information quality positively influences the flow experience of cultural heritage museum digital humans.

Aesthetic experience

Aesthetic experience fundamentally represents an evaluative process—a judgement of value[50]—that "creates new value in what we see and feel" [51]. Digital humans, as carefully designed aesthetic objects, activate this evaluative process through their distinctive interactive mechanisms. By integrating visual, linguistic, and behavioural cues, they foster continuous perceptual and emotional appraisal during interaction.

Technology enhances aesthetic experience primarily by facilitating deeper, more personalised interactions between users and exhibits[52]. This leads to both cultural fulfilment and aesthetic pleasure[53–55]. Effective aesthetic design captures users' attention and strengthens overall experience[56,57], which in turn improves perceptions of usefulness and usability[58]. Furthermore, in museum settings, design aesthetics correlate positively with behavioural intention[59,60], suggesting that aesthetic experience transforms into intrinsic motivation sustaining long-term interaction. Research also notes that the difficulty of understanding heritage content often stems from a lack of emotional activation mechanisms akin to those in cinema and literature[61]. Although multiple dimensions of immersion affect satisfaction, aesthetic experience exerts the strongest overall impact[62]. Hence, aesthetic experience not only arouses interest and appeal but also represents a deep evaluative awareness—"not perception of the external world, but awareness of our judgements about what matters to us" [51].

H6: Aesthetic experience positively influences the perceived usefulness of cultural heritage museum digital humans.

H7: Aesthetic experience positively influences perceived ease of use of cultural heritage museum digital humans.

H8: Aesthetic experience positively influences the flow experience of cultural heritage museum digital humans.

Perceived usefulness and perceived ease of use

Perceived usefulness (PU)—defined in this study as the extent to which users believe that digital humans facilitate their understanding of cultural heritage values—and perceived ease of use (PEOU)—the degree to which users perceive the system as easy to operate—constitute the core constructs of the Technology Acceptance Model (Davis[13]). Existing studies have demonstrated that PEOU significantly enhances PU[63–65], and that lower operational difficulty improves users' efficiency in obtaining information[66]. In immersive technology contexts such as virtual reality (VR) and virtual tourism, PEOU has been identified as a critical antecedent promoting users' entry into a flow state[67,68]. Moreover, the central conclusions of the TAM literature indicate that both PU and PEOU directly drive users' behavioural intention and are key predictors of continued usage[26,69].

H9: Perceived ease of use positively influences perceived usefulness of cultural heritage museum digital humans.

H10: Perceived ease of use positively influences the flow experience of cultural heritage museum digital humans.

H11: Perceived usefulness significantly influences users' usage intention.

H12: Perceived ease of use significantly influences users' usage intention.

Flow Experience

Flow denotes an optimal state of focused immersion accompanied by excitement and satisfaction[70]. In the case of museum digital humans, flow experience serves as a key indicator of narrative and interactional effectiveness. Through perceptibility and adaptive interaction, digital humans dynamically modulate users' engagement depth, creating a cyclical structure of challenge, feedback, and enjoyment that sustains flow[71]. This mechanism fosters continuous emotional and cognitive involvement, highlighting the unique strength of digital-human technologies in eliciting deep participation. Studies show that flow links high-quality user experience with sustained behavioural intention[72–74]. On one hand, the flow attribute of technology directly affects intention, thereby influencing acceptance and appreciation[75]; on the other, integrating flow into TAM enhances explanatory power[67]. Flow thus represents a critical emotional mechanism transcending utilitarian cognition. In other words, it captures not only immersion and enjoyment but also users' deep emotional resonance with digital humans. Accordingly, in this study, flow is conceptualised as the final output of the affective pathway and a high-order emotional state within the cognitive–affective dual-path model.

H13: Flow experience significantly influences users' usage intention.

Based on previous research, this study modifies the Technology Acceptance Model using empirical survey data, retaining perceived usefulness, perceived ease of use, and behavioural intention in TAM, while adding four external variables: information richness, information quality,

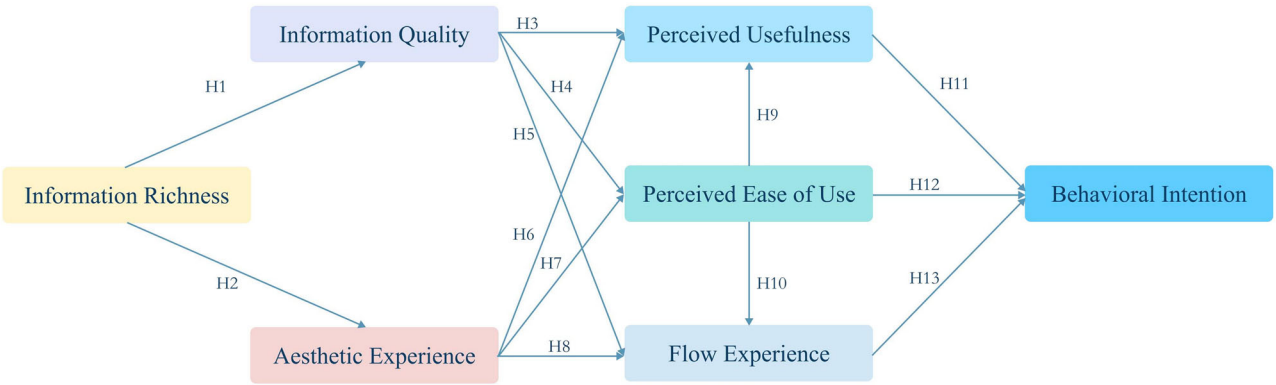

Fig. 2 | Research model of user intention of digital people in cultural heritage museums. The framework extends the original TAM by incorporating external factors related to information presentation and aesthetic experience to account for user intention in digital-human contexts.

Table 1 | Setting of scale variables and items

Variables	Constructs	Indicator contents	Source
Information Richness (IR)	IR1	The digital human presents information about the cultural heritage's construction history, cultural anecdotes, and artistic techniques.	63 86 87
	IR2	The digital human conveys implicit cultural memories and values embedded in the heritage.	
	IR3	The digital human's interpretation enables me to perceive social contexts and aesthetic pursuits embodied in the heritage.	
Information Quality (IQ)	IQ1	The interface design aligns cohesively with the information presented by the digital human.	88 89
	IQ2	Cultural heritage information delivered by the digital human is clear and vivid.	
	IQ3	Information conveyed by the digital human is coherent and logically structured.	
Aesthetic Experience (AE)	AE1	The visual design (colours, layout) of the digital human's interface is aesthetically pleasing.	90
	AE2	The digital human's expressive form and spirit evoke my associations with cultural heritage.	
	AE3	The digital human's dynamic expressions effectively guide my attention.	
Perceived Usefulness (PU)	PU1	The digital human's interpretation helps me acquire desired knowledge or information.	91 92
	PU2	The digital human deepens my understanding of cultural information.	
	PU3	Using the digital human is an effective method for communicating cultural heritage value.	
Perceived Ease of Use (PEOU)	PEOU1	Interacting with the digital human is intuitive and understandable.	91 92
	PEOU2	Obtaining information through the digital human is efficient.	
	PEOU3	Interaction with the digital human requires minimal cognitive effort.	
Flow Experience (FE)	FE1	I am intensely absorbed during the experience; nothing distracts me.	82 93
	FE2	I focus deeply and scarcely notice surrounding distractions.	
	FE3	I concentrate fully and experience time distortion.	
Behavioural Intention (BI)	BI1	I prefer using digital humans to access cultural heritage information.	94
	BI2	I will recommend digital humans to others for cultural heritage engagement.	
	BI3	I prioritize digital humans for heritage learning experiences.	
	BI4	I intend to explore more digital human applications in cultural heritage interpretation.	

aesthetic experience—to reflect the unique attributes of digital-human technology in cultural-heritage contexts (Fig. 2).

Survey design

This research employed a questionnaire survey to investigate factors influencing visitors' engagement with museum digital humans. The instrument comprised two sections: (1) demographic information (name, gender, education, occupation, prior experience, and familiarity with digital humans) and (2) seven latent constructs—information richness, information quality, aesthetic experience, perceived usefulness, perceived ease of use, flow experience, and behavioural intention—measured on a seven-point Likert scale (1= Strongly Disagree,7= Strongly Agree).

To ensure contextual suitability and content validity, expert evaluation and pilot testing preceded formal distribution. Five specialists in heritage communication, information systems, and psychology reviewed the items for semantic appropriateness, ensuring that adapted measures from established scales captured the study context accurately. A pilot test with fifty university students assessed comprehension, completion time, and logical flow, leading to minor wording refinements. Preliminary reliability analysis yielded Cronbach's $\alpha > 0.78$ for all constructs, indicating satisfactory internal consistency[76]. Table 1 presents the final constructs and measurement items.

Data collection

This study strictly adhered to ethical research standards. Questionnaires were administered anonymously, and no personally identifiable information was collected. Participants could withdraw at any point without consequence, and all data were used solely for academic purposes, ensuring no adverse impact on any individual.

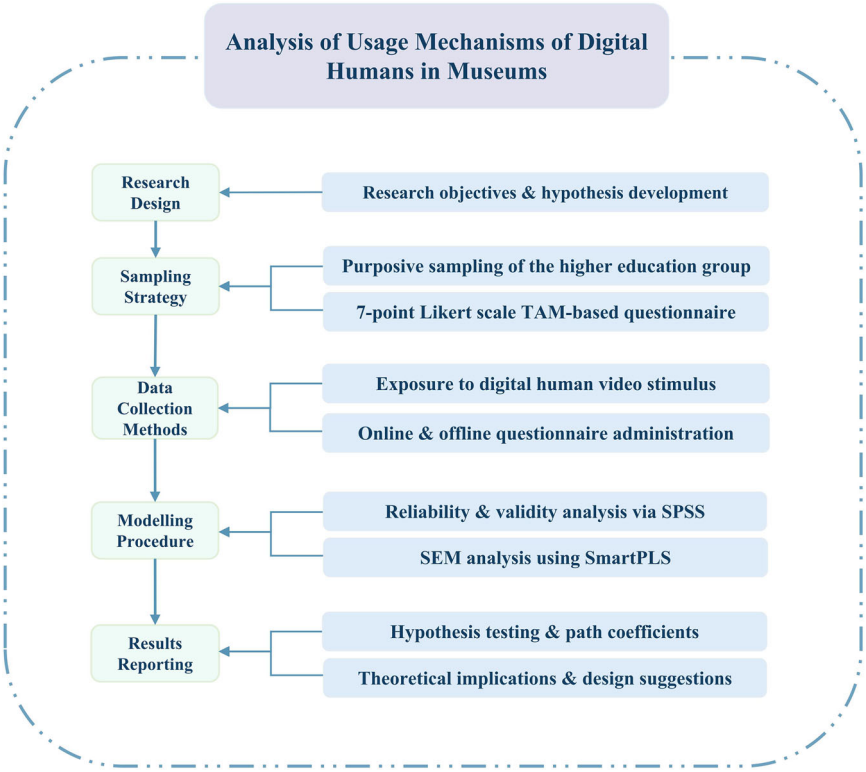

Fig. 3 | Research procedure for analyzing the usage mechanisms of digital humans in museums. The workflow progresses from research design and hypothesis development to data collection using digital-human stimuli and questionnaire surveys, followed by structural equation modelling and result reporting.

Table 2 | Descriptive statistics of the survey samples

Demographic Factors	Categories	Frequency	Percentage (%)
Gender	Male	100	37.7%
	Female	165	62.3%
Age	<18	2	0.75%
	18-25	218	82.26%
	26-30	28	10.57%
	31-40	10	3.77%
	>41	7	2.64%
Academic Level	High school and below	5	1.89%
	Junior college	13	4.91%
	Bachelor student	145	54.72%
	Master student	102	38.49%
Total		265	100%

As illustrated in Fig. 3, data collection employed a mixed strategy combining online and offline snowball sampling. Online surveys were managed through the professional platform "Wenjuanxing" and disseminated via WeChat social networks, while offline surveys were conducted on university campuses to reach the core demographic target. Although this approach effectively accessed relevant respondents, the homogeneity of the sample imposes certain limitations on the generalisability of findings to broader museum audiences, such as elderly visitors or international tourists.

To ensure the credibility and validity of collected data, participants were required to watch a 2–3-minute video of digital humans interpreting cultural heritage before completing the questionnaire. The experimental stimulus featured the "Ai Look Cultural Relics" short video series by the National Museum's digital human Ai Wenwen. Following the video, a content-screening question (e.g., "Which artefact was primarily featured in the video?") ensured that only attentive participants continued. Those answering incorrectly were automatically excluded by the system. The formal survey was conducted from early to late October 2024, yielding 319 responses. After applying multiple screening criteria—excluding submissions with excessively short completion times, repetitive answers, or duplicate entries—265 valid questionnaires were retained, representing an effective rate of 83.07%. The final sample size exceeded 10 times the number of measurement items (22), thus satisfying Structural Equation Modelling (SEM) requirements. The survey respondents primarily comprised Chinese higher education cohorts, including undergraduate, master's, and doctoral students. Given the experimental materials' origin from the National Museum of China's digital human exhibition, Chinese participants could generate deeper cognitive resonance and immersion based on cultural identification.

Respondent background

Descriptive statistical analysis using SPSS 26.0 is presented in Table 2. The sample was predominantly female (62.3%). Regarding education level, most respondents held a bachelor's degree (54.72%) or a postgraduate qualification (39.49%). In terms of age, the majority (82.26%) were between 18 and 25 years old.

These characteristics reflect emerging trends in digital-cultural participation: young, well-educated "digital natives" whose media habits align closely with the experimental design and who are generally comfortable using digital technology to acquire information (Zollo et al. [77]).

Descriptive statistics were generated using SPSS 26.0, with detailed sample characteristics shown in Table 2. Among respondents, females constituted a larger proportion (62.3%). Regarding educational attainment, most participants held bachelor's degrees (54.72%) or higher qualifications (39.49%). In terms of age distribution, subjects were predominantly concentrated in the 18-25 age range, accounting for 82.26% of all participants. These sample characteristics reflect emerging trends in digital cultural dissemination: young, highly educated groups—as "digital natives"—exhibit strong alignment between their media usage habits and experimental design parameters, and this demographic demonstrates greater familiarity with digital technologies for information acquisition[77].

Table 3 | Measurement model assessment

Constructs	Items	Loadings	Cronbach α	CR[a]	AVE[b]
IR	IR1	0.883	0.833	0.900	0.749
	IR2	0.857			
	IR3	0.857			
IQ	IQ1	0.836	0.801	0.882	0.714
	IQ2	0.868			
	IQ3	0.831			
AE	AE1	0.812	0.797	0.881	0.711
	AE2	0.857			
	AE3	0.860			
PU	PU1	0.861	0.793	0.878	0.706
	PU2	0.847			
	PU3	0.811			
PEOU	PEOU1	0.878	0.778	0.871	0.693
	PEOU2	0.794			
	PEOU3	0.823			
FE	FE1	0.865	0.867	0.918	0.733
	FE2	0.899			
	FE3	0.901			
BI	BI1	0.880	0.878	0.916	0.733
	BI2	0.851			
	BI3	0.825			
	BI4	0.868			

[a]*CR* Composite Reliability; [b] *AVE* Average Variance Extracted.

Table 4 | Analysis of discriminant validity

Constructs	IR	IQ	AE	PU	PEOU	FE	BI
IR	0.866						
IQ	0.563	0.845					
AE	0.526	0.490	0.843				
PU	0.521	0.541	0.517	0.840			
PEOU	0.491	0.558	0.487	0.581	0.832		
FE	0.399	0.426	0.492	0.465	0.574	0.889	
BI	0.561	0.494	0.598	0.545	0.611	0.679	0.856

Results

Reliability, convergent, and discriminant validity

Prior to hypothesis testing, multicollinearity diagnostics were performed on predictor variables in the structural model. Results show all VIF values were below the critical threshold of 5[78]. Therefore, collinearity does not affect the reliability and validity of the measures.

All measurement indicators met accepted reliability and validity thresholds (factor loadings ≥ 0.70; Cronbach's α and composite reliability ≥ 0.70; Average Variance Extracted (AVE) ≥ 0.50). Discriminant validity was confirmed following Fornell and Larcker[76]. As shown in Tables 3 and 4, the scales used in this study demonstrated stable reliability, strong internal consistency, and satisfactory discriminant validity.

Structural model and path coefficients

Structural paths are summarised in Fig. 4 and Table 5. Information richness significantly enhanced perceived information quality ($\beta = 0.563$, $p < 0.01$) and aesthetic experience ($\beta = 0.526$, $p < 0.01$). Information quality improved perceived usefulness ($\beta = 0.239$, $p < 0.01$) and ease of use ($\beta = 0.420$, $p < 0.01$), but did not significantly affect flow ($\beta = 0.072$, $p > 0.05$). Aesthetic experience positively influenced usefulness ($\beta = 0.239$, $p < 0.01$), ease of use ($\beta = 0.281$, $p < 0.01$), and flow ($\beta = 0.258$, $p < 0.01$). Ease of use, in turn, improved usefulness ($\beta = 0.332$, $p < 0.01$) and flow ($\beta = 0.408$, $p < 0.01$). Usefulness ($\beta = 0.198$, $p < 0.01$), ease of use ($\beta = 0.237$, $p < 0.01$), and flow ($\beta = 0.451$, $p < 0.01$) all significantly predicted behavioural intention.

Overall, the model explained a moderate proportion of variance (PU $R^2 = 0.439$; PEOU $R^2 = 0.366$; FE $R^2 = 0.385$), indicating that most direct and mediating paths achieved statistical significance.

Model comparison and incremental explanatory power

To verify the theoretical contribution, three nested models were compared (Table 6).

The baseline simplified TAM model (Model 1) containing only perceived usefulness and ease of use explained 42.4% of the variance in usage intention ($R^2 = 0.424$). Unexpectedly, adding system-information characteristics (information richness and information quality) to form an enhanced TAM (Model 2) resulted in negligible improvement ($R^2 = 0.425$; $f^2 \approx 0$). This indicates that in the specific context of museum digital humans, the traditional "system characteristics → cognitive judgement → intention" pathway contributes minimally. Information features alone are insufficient to drive behavioural intention.

By contrast, the introduction of aesthetic and flow experiences in the dual-path extended model (Model 3) produced a substantial increase in explanatory power ($R^2 = 0.554$), with an effect size $f^2 = 0.29$—approaching a large effect according to Cohen's[79] guidelines. This clearly demonstrates that the emotional–immersive pathway proposed in this study constitutes the primary mechanism driving user acceptance of digital humans.

Q2 predictive relevance

Prediction errors are evaluated through Q^2 values, and $Q^2 > 0$ indicates predictive relevance for specific variables. Q^2 values for IQ (0.311), AE (0.271), PU (0.250), PEOU (0.225), and FE (0.150) demonstrate that, based on Q^2-driven predictive power assessment, all endogenous latent variables achieve moderate-to-high predictive levels.

Discussion

By introducing a cognitive–affective dual-path model, this study extends the classical Technology Acceptance Model (TAM) within the context of digital-human applications for cultural-heritage interpretation and proposes several theoretical contributions.

First, the findings confirm the central importance of the affective pathway in technology acceptance for cultural-heritage experiences, thereby expanding the theoretical boundary of TAM. The pathway driven by aesthetic experience and mediated through flow experience[32] demonstrates explanatory power equal to, and in some cases exceeding, that of the traditional cognitive route. This evidence that users' acceptance of museum digital humans is not grounded solely in utilitarian "instrumental rationality" but relies heavily on emotional resonance and immersion. In this sense, TAM evolves from an efficiency model to an experience-acceptance model appropriate for "affective media", deepening our understanding of technology-adoption behaviour in cultural contexts.

Secondly, the study not only validates the relevance of information richness within heritage environments but also positions aesthetic experience and flow experience as core explanatory variables in digital-curation studies. Results show that information richness enhances both information quality and aesthetic experience, which together indirectly reinforce usage intention through flow. This demonstrates the importance of multimodal narrative structures in digital-heritage presentation: layered, well-organised information does more than support knowledge transfer[40]; it creates a progressive chain linking cognitive load, aesthetic pleasure, and immersive flow. Unlike traditional displays, digital humans synchronise sensory stimulation and cultural value through speech, expression, and gesture, merging technological function with heritage significance within the user's experience.

Thirdly, the study identifies a distinct mechanism by which digital humans achieve effective cultural-heritage interpretation. Both information

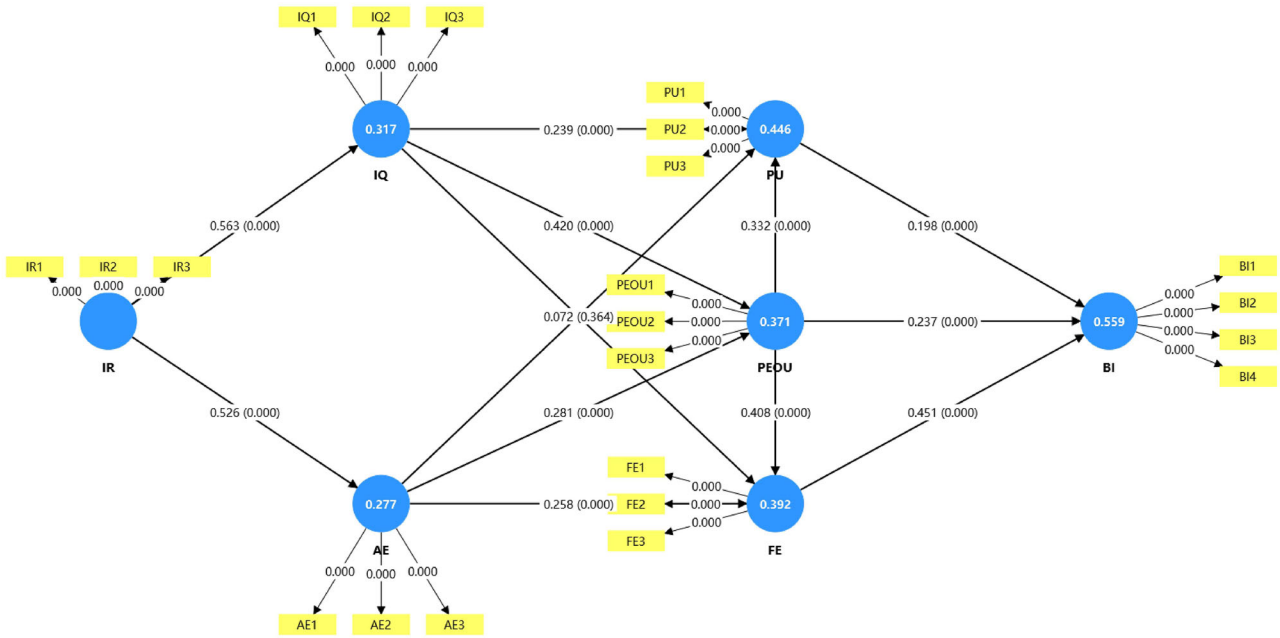

Fig. 4 | Diagram of structural equation model. The diagram presents the estimated relationships among cognitive, experiential, and affective factors influencing behavioural intention.

Table 5 | Results of hypotheses testing

Hypotheses	Relationships	Path Coefficients	Sample Mean	STDEV	T Values	*p*	Decision
H1	IR → IQ	0.563	0.563	0.044	12.804	0.000	Supported
H2	IR → AE	0.526	0.525	0.054	9.729	0.000	Supported
H3	IQ → PU	0.239	0.241	0.063	3.814	0.000	Supported
H4	IQ → PEOU	0.420	0.419	0.053	7.881	0.000	Supported
H5	IQ → FE	0.072	0.073	0.080	0.907	0.364	Not Supported
H6	AE → PU	0.239	0.240	0.054	4.452	0.000	Supported
H7	AE → PEOU	0.281	0.283	0.064	4.389	0.000	Supported
H8	AE → FE	0.258	0.258	0.065	3.988	0.000	Supported
H9	PEOU → PU	0.332	0.329	0.059	5.613	0.000	Supported
H10	PEOU → FE	0.408	0.408	0.071	5.793	0.000	Supported
H11	PU → BI	0.198	0.197	0.054	3.656	0.000	Supported
H12	PEOU → BI	0.237	0.238	0.065	3.638	0.000	Supported
H13	FE → BI	0.451	0.451	0.061	7.363	0.000	Supported

Table 6 | Comparison of nested models and analysis of incremental explanatory power

Model	Description	Included Constructs	R^2 for Behavioural Intention	Effect Size (f^2)	Interpretation
Model 1: Simplified TAM	Core TAM pathways only	PU, PEOU,	0.424	–	Baseline
Model 2: Enhanced TAM	Adds system information features	PU, PEOU, IR, IQ	0.425	0.02(compared with Model1)	Traditional path ineffective; system features add minimal explanatory power
Model 3: Extended Dual-Path Model | Adds emotional–immersive pathway (AE, FE)	Adds emotional–immersive pathway (AE, FE)	PU, PEOU, IR, IQ, AE, FE	0.554	0.29(compared with Model2)	Emotional–immersive path dominant; large incremental explanatory power

*The interpretation of the f^2 effect size follows the criteria of (Cohen, 1988)[79]: 0.02 (small), 0.15 (medium), 0.35 (large).

quality and aesthetic experience enhance perceived usefulness and ease of use, yet only aesthetic experience—by evoking a state of timeless immersion—significantly predicts sustained usage intention. Hence, users' deep engagement arises primarily from aesthetic enjoyment and emotional involvement[80] rather than rational evaluation alone. High-quality information and visual design jointly form a trust–pleasure–immersion loop, in which credibility ("hard power") underpins emotional attraction and narrative aesthetics ("soft power") ignite immersion. Contrary to earlier perspectives that regarded

technological experience merely as a delivery mechanism, this research foregrounds a narrative-interpretive shift: through visual aesthetics and multimodal semiotic translation, digital humans re-embody heritage value in emotionally expressive and experientially rich ways.

From a practical standpoint, the findings yield multiple insights for museum curation and heritage conservation: Information richness should be treated as a foundational element of cultural interpretation rather than a purely technical metric. Digital-human narratives must align their informational depth with the cultural context to avoid "technological excess" and cognitive dissonance. Aesthetic design must serve the perception of cultural meaning. Narrative context, visual composition, and interaction tempo should guide visitors towards understanding through pleasure. Flow generation is not an automatic outcome of technology but a composite effect of heritage content, storytelling strategy, and user psychology[33]. Consequently, digital-human adoption in curatorial practice should prioritise cultural interpretation, transforming immersive experience into renewed cognition and long-term memory of heritage significance.

Within the field of cultural-heritage practice, the development trajectory of museum digital humans follows two complementary orientations—cultural dissemination and cultural safeguarding. The former, represented by Ai Wenwen at the National Museum of China, leverages multimodal storytelling and intelligent interaction to strengthen knowledge transmission and emotional resonance, contributing to the digital articulation of national identity. The latter, exemplified by the Dunhuang Academy's digital human Jiayao, employs virtual reconstruction and immersive experience to enable public interpretation and non-invasive conservation, extending exhibition and educational functions. Together, they constitute a dual-path model of practice in which dissemination reinforces cognitive engagement, and safeguarding extends cultural memory.

Nevertheless, while digital-human technologies effectively alleviate physical-site pressures, their substitutive experience carries limitations: excessive reliance on digital mediation may weaken perceptions of heritage authenticity[81]. Thus, design strategies must balance perceptible immersion with respectful authenticity, ensuring that technology operates as an extension—not a replacement—of heritage protection, achieving sustainable synergy between digital communication and cultural conservation.

A noteworthy outcome of this study is the non-significant relationship between information quality and flow experience(H5). Consistent with previous findings that flow depends more on interactivity, situational perception, enjoyment, and playfulness[82–84], this does not imply that information quality is unimportant. Rather, it reveals that within the experiential technology of museum digital humans, the cognitive and affective pathways operate under distinct mechanisms and boundaries.

Flow represents an intense, time-suspending state driven primarily by immediate emotional resonance, sustained attention, and appropriately balanced challenge[70,71]. Information quality, although crucial for trust and rational evaluation, functions as a background enabler rather than a trigger. Even perfectly accurate information cannot elicit flow if presented through dull or clumsy interaction.

This finding substantiates the discriminant validity of the dual-path model: information quality initiates the cognitive route (PU, PEOU), whereas aesthetic experience propels the affective-immersion route (FE). The absence of support for H5 thus clarifies that success with digital humans depends on balancing rational credibility with emotional allure.

Although this study constructs and validates an acceptance model for cultural-heritage digital humans incorporating information richness, information quality, aesthetic experience, and flow, several limitations and opportunities remain.

The dataset consisted almost entirely of Chinese university students aged 18–25 (82%), a group influential in digital-culture consumption yet distinct from other demographic segments in cultural identity, aesthetic preference, and technological familiarity. This restricts generalisability to broader visitor populations, particularly older or international audiences. Future research should adopt cross-age and cross-cultural sampling to test the model's universality and cultural adaptability.

This study employed a quantitative design combining questionnaires and SEM analysis, which effectively identifies structural relationships but cannot capture the dynamic evolution of psychological mechanisms such as flow and aesthetic perception. Subsequent studies may integrate experimental, eye-tracking, EEG, or behavioural-tracking approaches to objectively record users' perceptual–cognitive–affective processes within immersive heritage settings. Longitudinal or situational designs could also examine how flow varies with interaction duration, task complexity, or narrative pacing.

Although the model integrates information richness and the "what-is-beautiful-is-usable" principle[85]alongside flow theory, it remains perception-centred. Additional latent variables—such as cultural identity, perceived authenticity, narrative engagement, and emotional resonance—may further enrich explanatory depth.

This research focused on the National Museum's Ai Wenwen as a representative case. While this ensured analytical depth, digital humans associated with different thematic categories—archaeological sites, folk art, or regional heritage—may involve distinct motivational structures. Future comparative work could develop typological models elucidating structural differences across cultural domains.

Digital-human technology, through embodied and personified narration, offers an animated interpretive pathway for heritage: transforming static history into dynamic storytelling that stimulates emotional resonance and cognitive depth. Future investigations may extend the framework theoretically by incorporating cultural identity and social presence as mediators, or empirically by exploring how digital-human personality, narrative style, and design specificity influence engagement.

Moreover, cross-contextual studies comparing archaeological, artistic, and intangible heritage applications could examine the contextual flexibility of the model. As multimodal interaction and affective computing advance, scholars should continue to explore how digital humans employ language, expression, and narrative rhythm to realise personalised cultural communication, facilitating the transition from informational experience to cultural resonance.

Data availability

The original contributions presented in this study are included in the article and supplementary material. Further inquiries can be directed to the corresponding author(s).

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Author contributions

For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used "Conceptualization, J.M. and C.C.; methodology, H.C. and C.Y.; software, Huizhen Chen; validation, C.Y., C.C. and J.M.; formal analysis, C.Y.; investigation, H.C.; resources, J.M.; data curation, Z.W.; writing—original draft preparation, H.C. and C.Y.; writing—review and editing, H.C.; visualization, C.Y.; supervision, C.C.; project administration, C.C.; funding acquisition, C.C. All authors have read and agreed to the published version of the manuscript." Please turn to the CRediT taxonomy for the term explanation. Authorship must be limited to those who have contributed substantially to the work reported.

Competing interests

The authors declare no competing interests.

Additional information

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